

EMMANUEL PADDY ADAMS

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine learning engineer with a BSc in Statistics with Computer Science and hands-on experience building regression and classification models end to end: sourcing and joining messy data (including APIs), engineering features, validating honestly and packaging models as reusable scikit-learn pipelines. Also delivers SQL analysis and Power BI dashboards for non-technical stakeholders. Cohort 1 graduate of DataBloom Africa; data analyst intern in regulated financial services.

TECHNICAL SKILLS

Languages: Python, SQL

Machine learning: scikit-learn (pipelines, ColumnTransformer), OLS / Lasso / Ridge, Random Forest, Gradient Boosting, XGBoost, SMOTE, cross-validation, feature engineering

Data & analysis: Pandas, NumPy, EDA, statistical analysis, REST APIs (NASA POWER), geocoding (geopy)

Visualisation & BI: Power BI (DAX, drill-through dashboards), Matplotlib, Seaborn, Excel (pivot tables, advanced formulas)

Databases & tools: MySQL, PostgreSQL, Git, GitHub, Jupyter, VS Code, Figma

PROFESSIONAL EXPERIENCE

Unique Insurance, Accra / *Data Analyst Intern*

Nov 2025 – Dec 2025

- Validated and audited insurance datasets for 3 internal teams, flagging inconsistencies across 1,000+ records to reduce reporting discrepancies.
- Built ad-hoc reports and templates adopted by the audit team for monthly use, applying Excel controls (VLOOKUP, SUMIF) in a compliance-driven environment.

BCG · Forage virtual programme / *Data Analyst Simulation*

Mar 2026

- Analysed multi-channel advertising campaign data to evaluate conversion rates and ROI across audience segments.
- Identified the highest-performing channels and produced a recommendation deck with data-backed marketing strategy.

SELECTED PROJECTS

Ghana Maize Yield Prediction / *Machine Learning*

Python · scikit-learn · NASA POWER API

- Built an end-to-end pipeline predicting district-level maize yield, joining 1,681 district-year yield records with daily NASA POWER weather (geocoded per district) and regional soil types.
- Engineered seasonal weather aggregates, a yield lag and pest/policy indicators; trained a Random Forest + Gradient Boosting ensemble in a saved scikit-learn pipeline (R^2 0.95 on a random hold-out; now re-validating with a time-based split).

Used Car Price Prediction / *Machine Learning*

Python · scikit-learn

- Modelled prices for ~11,900 vehicles; one-hot encoding with rare-category grouping raised test R^2 from 0.57 (numeric baseline) to 0.82.
- Used cross-validated Lasso for feature selection (86 of 182 features kept); Ridge reached R^2 0.87 under 5-fold cross-validation.

Ghana Tech Ecosystem Survey 2025 / *Analytics*

Python · Power BI · DAX

- Cleaned survey responses from 286 tech professionals (43 questions reshaped into a 7,425-row table) and built a 4-page drill-through Power BI dashboard with 18 DAX measures on roles, skills gaps, AI adoption and ecosystem needs.

Blinkit Grocery Sales Analysis / *BI*

SQL · Power BI

- Cleaned 8,523 item records across 10 outlets in SQL (merged inconsistent category labels) and built a Power BI dashboard of sales KPIs by item type, outlet tier and fat content.

Data Science Job Market Analysis / *BI*

Excel

- Benchmarked salaries for 2,000+ postings across 10+ job titles and 5 industries; found a 35% junior–senior pay gap and Excel, SQL and Python as the top-requested skills.

EDUCATION

University of Ghana / *BSc Statistics with Computer Science*

2026

Key courses: Advanced Linear Regression, Machine Learning, Database Systems & Management, Theory of Sampling.

TRAINING & CERTIFICATIONS

DataBloom Africa / *Data Analysis Programme, Cohort 1 (certified)*

2025

ALX Africa / *Professional Foundations & Data Analytics (certified)*

2024 – 2025

Cognitive Class (IBM) / *Data Analysis with Python*

2025

LEADERSHIP

- Team Lead, Data Analysis Hackathon:** led a 4-person team to deliver a complete analysis within 8 hours.
- Group Leader, ALX Professional Foundations:** led a Figma UI/UX prototype of a mental-wellness app for female students.